# Angry Wolves V2 Stability + Tuning Pass Report

Branch: `refresh/v2-barnyard-core`

Date: 2026-04-26

Scope: scoring stability, ghost/drop-lane readability, and iPhone audio reliability.

Status: implemented locally for review only. Nothing was committed, merged, deployed, pushed to `main`, or changed in GitHub Pages settings.

## Executive Summary

This focused pass addressed three urgent V2 issues:

- Scoring was too volatile because eggs used exponential doubling ( `2^eggCount` ) and could create runaway payouts.
- The landing ghost/drop lane was too bright and visually competed with the active falling tetrad.
- iPhone audio could appear broken because missing SFX volume storage was being read as `0%`.

The fix keeps V2 fun and readable while preserving the legacy path behind `?v1=1`.

## Files Changed

- `game.js`: V2 scoring rebalance, quieter ghost/drop lane, audio reset/debug/test flow, V2 leaderboard tagging.
- `index.html`: Settings version/cache bust, SFX default display, Test Sound button, updated scoring help.
- `ROLLBACK_PLAN.md`: rollback notes for this pass.
- `refresh-assets/stability-tuning-pass/`: screenshot evidence.

## Scoring Audit

Before this pass, V2 scoring used:

- Base herd score: `herdSize`.
- Herd size bonus: Fibonacci-style bonus above the herd threshold.
- Egg bonus: each egg doubled the herd payout, so `1 egg = x2`, `2 eggs = x4`, `3 eggs = x8`.
- Turd penalty: each turd halved the herd payout.
- Chain bonus: separate Fibonacci-style bonus after clears.
- Mission bonus: separate reward-cashout bonus added at run end.
- Angry Wolves bonus: large marquee mission bonus, currently `+540` after tuning.

The most likely runaway source was verified: eggs were exponential.

## Scoring Rebalance

V2 now uses bounded, readable scoring:

- Herd base: `herdSize * 5`.
- Size bonus: `+8` per animal above the V2 herd threshold.
- Eggs: `+20%` per egg, capped at four eggs for a max `x1.8`.

- Turds/mud: `-18%` per turd, capped at three turds with a `0.55` payout floor.
- Chains: separate bounded flat combo bonus, capped at `+120`.
- Mission bonuses stay separate and are not multiplied by eggs/chains.

V2 leaderboard submissions are now tagged separately:

- `GAME_MODE = "v2-prototype"`
- `GAME_VERSION = "v0.35-v2-score-stable"`

This prevents the unstable prototype economy from blending with the normal public board.

## Sample Scoring Table

| Case | Before | After |
|---|---|---|
| 9-animal herd, no egg/turd | 11 | 45 |
| 12-animal herd, no egg/turd | 20 | 84 |
| 9-animal herd, 1 egg | 22 | 54 |
| 9-animal herd, 3 eggs | 88 | 72 |
| 12-animal herd, 2 eggs, 1 turd | 40 | 96 |
| x2 chain bonus | 5 | 18 |
| x3 chain bonus | 8 | 45 |
| normal mission completion | 80 | 80 |
| Angry Wolves completion | 540 | 540 |

## Ghost / Landing Indicator Tuning

The active falling tetrad should be more visible than its landing hint. The ghost and drop lane were reduced to make the board calmer.

| Constant / Treatment | Before | After |
|---|---|---|
| Ghost token alpha | `0.38` | `0.24` |
| Ghost token base alpha | `0.72` | `0.52` |
| Ghost cell fill | `rgba(255,235,173,0.065)` | `rgba(234,222,176,0.035)` |
| Ghost cell stroke | `rgba(255,231,172,0.74)` | `rgba(238,226,184,0.42)` |
| Ghost cell line width | `0.028 * cell` | `0.018 * cell` |
| Drop lane bottom alpha | `0.16` | `0.055` |

Visual hierarchy after tuning:

- Active falling tetrad is primary.
- Settled animals remain clear.
- Ghost footprint is useful but quiet.
- Drop lane is subtle.
- Herd highlights remain separate from the ghost.

## Audio Findings And Fixes

The Settings screenshot from the prior mission review showed `SFX Volume` at `0%`. The bug was confirmed:

- The previous loader used `Number(localStorage.getItem("aw_sfx_volume"))`.
- When no value was stored, `localStorage.getItem(...)` returned `null`.
- `Number(null)` is `0`, so a missing preference became `0%`.

New defaults:

- `aw_sound = "1"`
- `aw_sfx_volume = "0.65"`
- `aw_goofy_animals = "1"`

New controls/debug tools:

- Settings includes a `Test Sound` button.
- `?audioReset=1` restores safe audio defaults.
- `?audioDebug=1` logs audio state transitions.
- `?debugScore=1` logs scoring samples and herd score breakdowns.

Important iPhone note:

- No silent-switch bypass was added.
- If iPhone silent mode is ON, silence remains correct.
- If silent mode is OFF and sound is enabled, the game now has better defaults and an explicit gesture-based Test Sound path.

## Validation

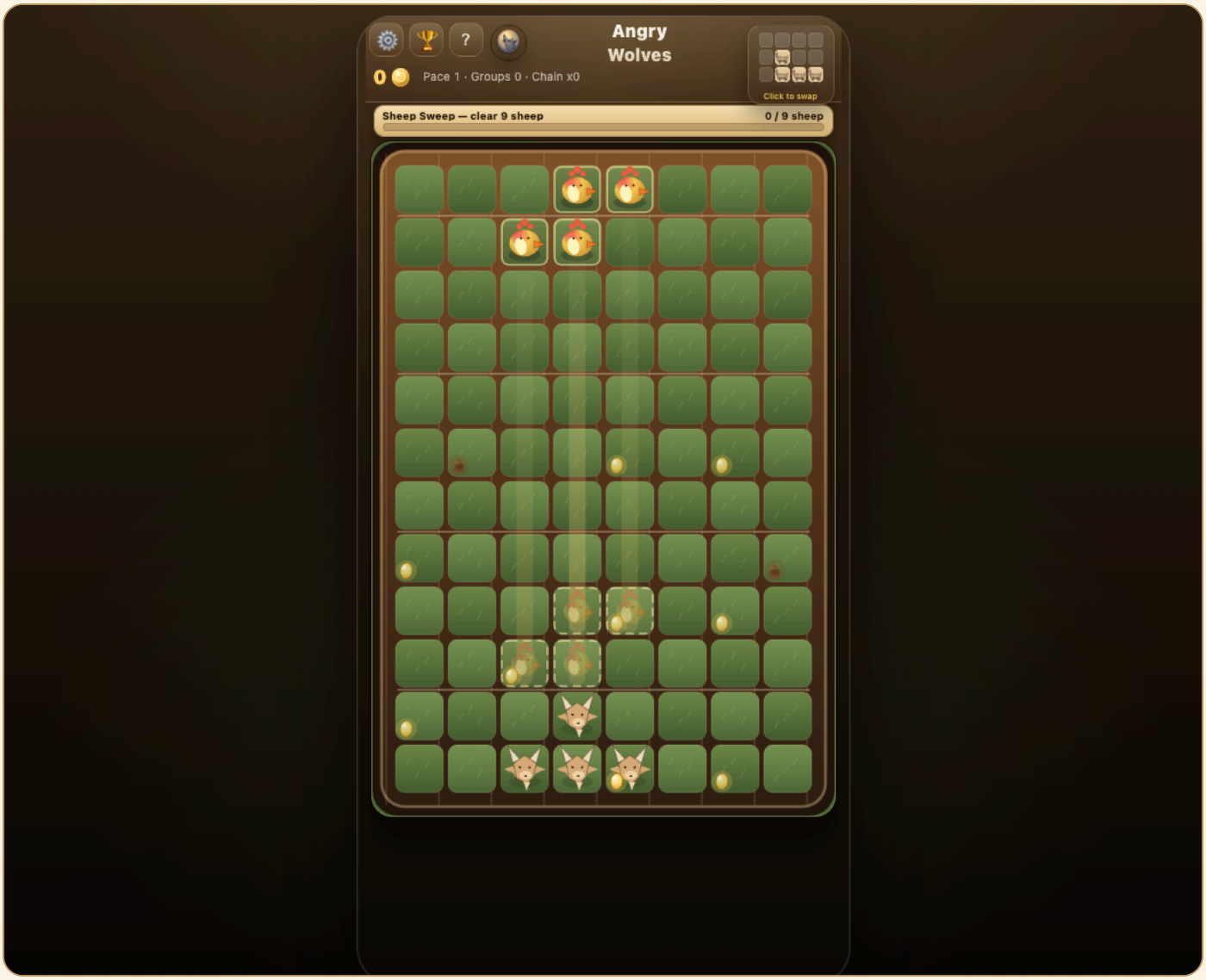
- `node --check game.js`: passed.
- `git diff --check`: passed.
- No `package.json`; this is a static repo with no npm scripts.

Local smartphone test URL used during the pass:

`http://192.168.0.30:4175/index.html?audioReset=1&audioDebug=1`

## Screenshots

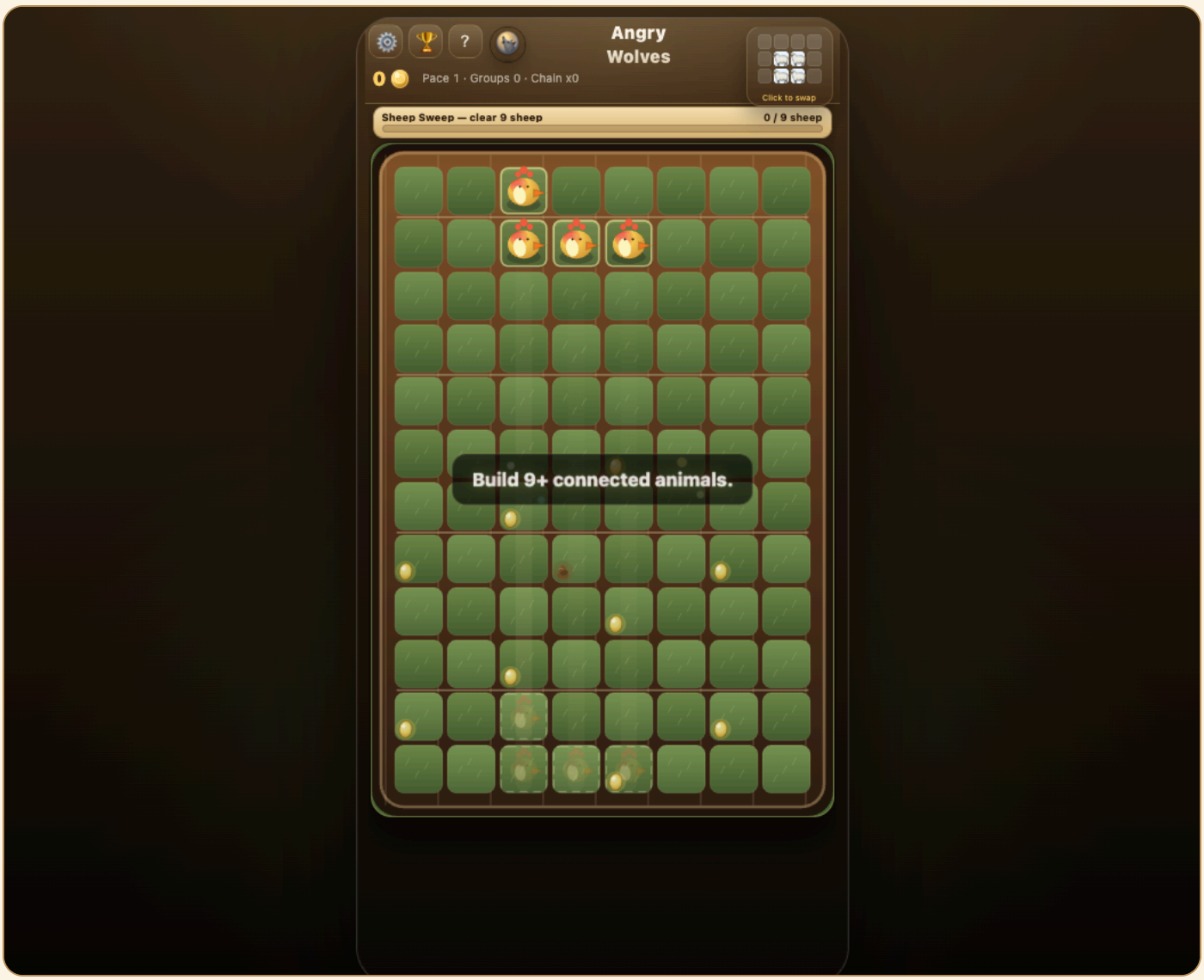
### Before: Ghost / Settings Baseline




Before ghost/settings baseline

This capture preserves the pre-pass state for comparison.

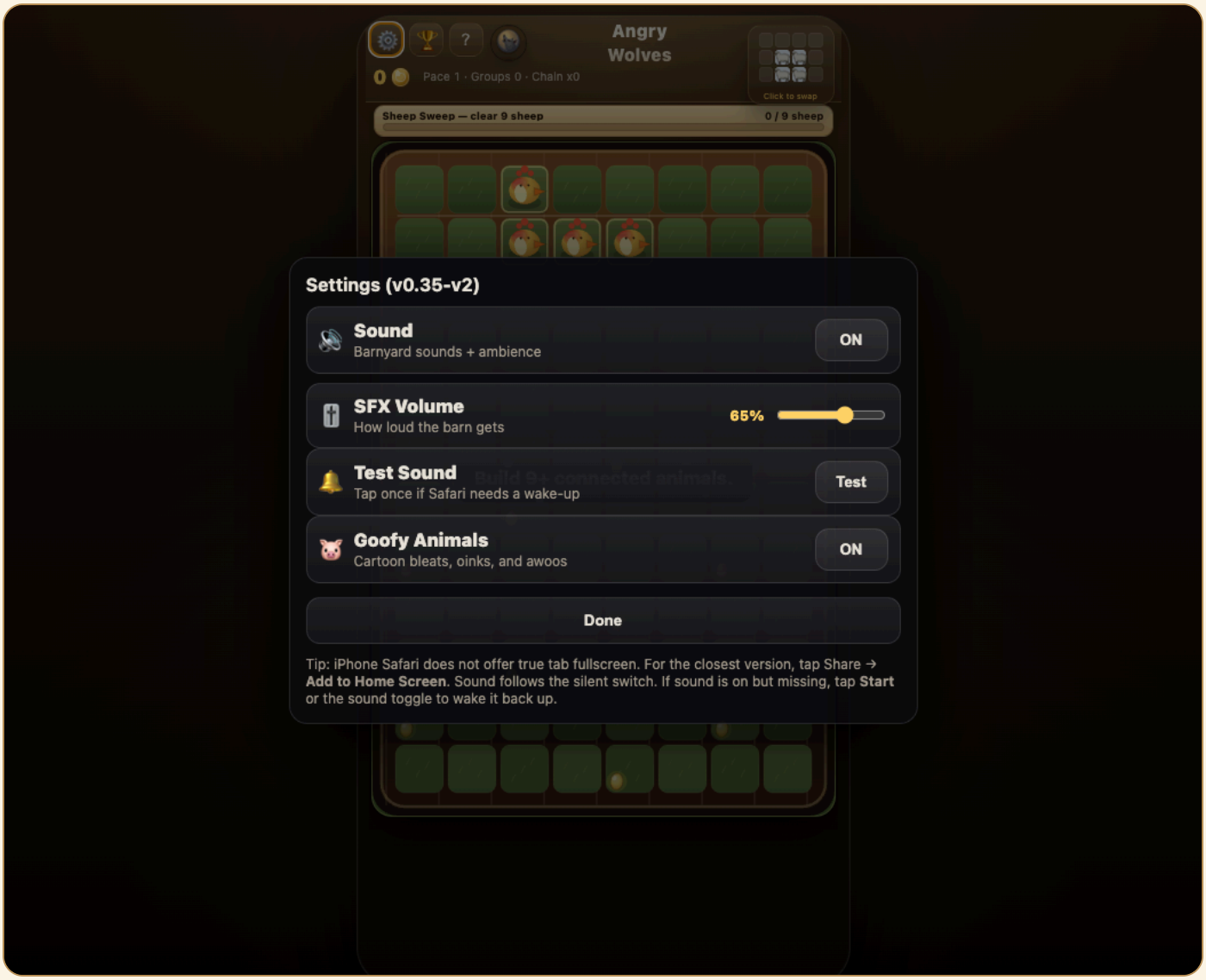
## After: Quieter Ghost / Drop Lane




After quieter ghost and drop lane

The ghost landing footprint is now much quieter than the active tetrad. The drop lane still indicates where the piece will land, but no longer reads as a bright board feature.

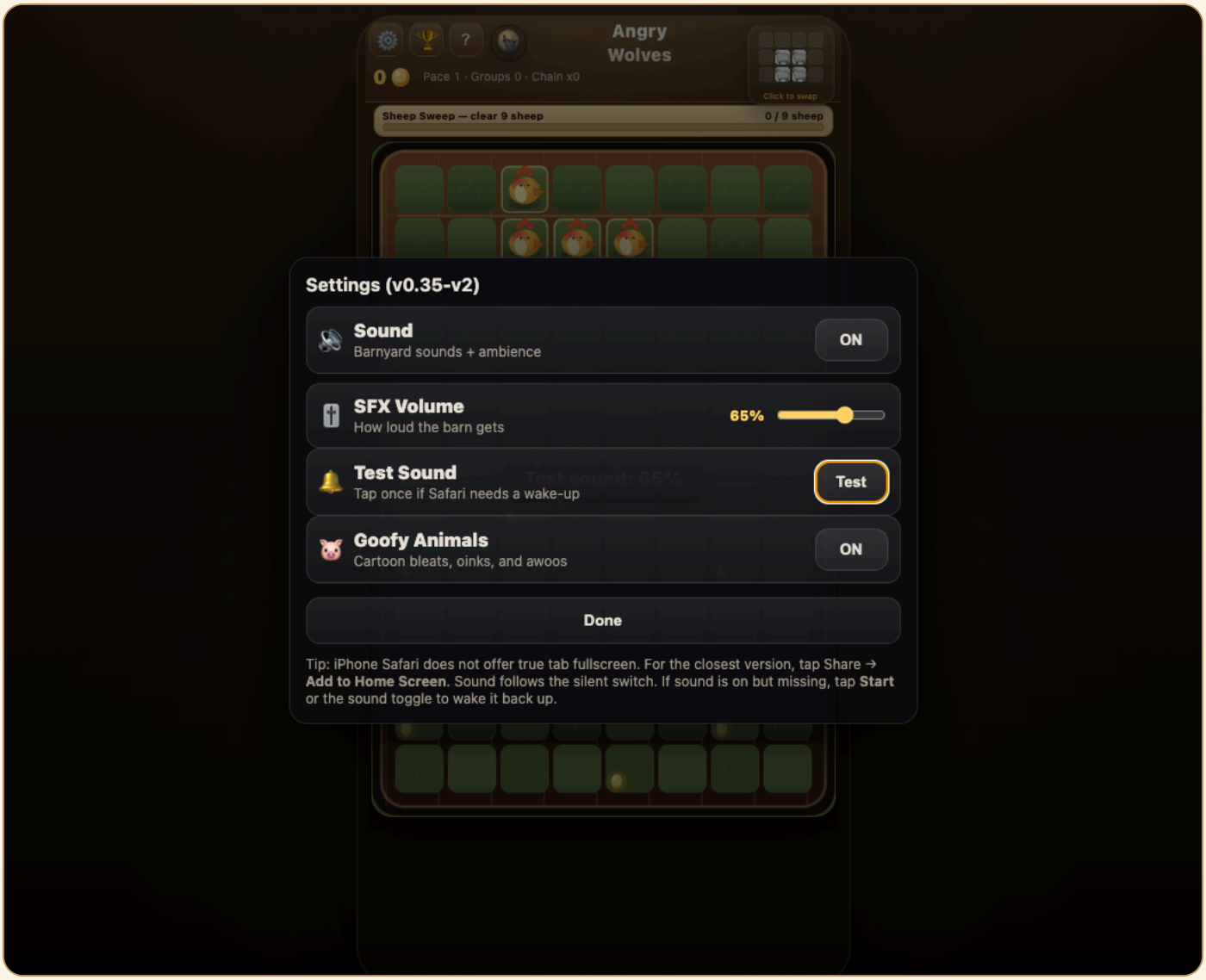
## After: Settings With Safe Audio Default




Settings with Test Sound and 65 percent SFX

After `?audioReset=1`, SFX volume correctly defaults to `65%` instead of `0%`.

### After: Test Sound Button




Test Sound button confirmation

The Settings panel now includes a native-looking `Test Sound` control. It attempts to resume audio from the button gesture and plays a small UI/pig cue if sound is enabled and volume is above zero.

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## Rollback Summary

The rollback plan was updated with targeted instructions:

- Revert only V2 scoring by restoring the old `chainBonusForDepth`, `herdSizeBonus`, and herd scoring block.
- Revert only ghost tuning by restoring the previous ghost/drop-lane constants and draw logic.
- Revert only audio settings/debug by removing `Test Sound`, `audioReset`, `audioDebug`, and the safe-default loader changes.
- Use `?v1=1` to bypass the V2 path entirely.

## Final Notes

This pass intentionally did not redesign missions, change GitHub Pages settings, merge, deploy, push to `main`, or commit. It keeps the branch reviewable and reversible while making the current V2 build more stable for phone testing.